Closing the low-q gap for the dimension formula: structural proofs at $(3, 2, 1^q)$ for $q \geq 1$ and $(4, 2, 1^q)$ for $q \geq 2$

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Abstract

The May-13 multi-corner dimension paper conjectured (with conditional proof modulo Phase A and Hoefsmit nonvanishing) the closed-form $\dim B^{(\lambda)}_{\ell+1} V_\lambda = f^{\lambda^{(\geq 2)}}$ in the stable regime. The existing structural-j-formula club (May-12 / May-20) covers $(3, 2, 1^q)$ at $n \geq 8$, i.e. $q \geq 3$, and $(4, 2, 1^q)$ at $n \geq 10$, i.e. $q \geq 4$.

This paper closes the low-q gap structurally:

Theorem A. $\dim B^{(\lambda)}_{\ell+1} V_\lambda = 2 = f^{(2,1)}$ for $\lambda = (3, 2, 1^q)$ at every $q \geq 1$.

Theorem B. $\dim B^{(\lambda)}_{\ell+1} V_\lambda = 3 = f^{(3,1)}$ for $\lambda = (4, 2, 1^q)$ at every $q \geq 2$.

Each proof is by the four-branch Phase A decomposition + commutation lemma, with the bad-pair and same-row vanishing arguments at $q \geq 2$ deriving from the night $T_1 = -1$ threshold $q_0(\widehat{\nu}) = |\widehat{\nu}| - 2\ell(\widehat{\nu}) + 2$ applied to each intermediate piece. The structural inputs reduce to hook j-formulas on $(2, 1^{q+1}), (3, 1^q), (3, 1^{q+1})$ and the two-column $(2, 2, 1^q)$ j-formula, all unconditional in the appropriate ranges.

The same template fails at $q = 1$ for $(4, 2, 1^1)$, where direct computation gives $\dim B = 6 \neq 3$. We dissect this non-stable failure explicitly: the bad-pair branch contributes 1 extra and the same-row branch contributes 2 extra, exactly accounting for the $6 - 3 = 3$ deficit. This is the only non-monotonicity in the $(4, 2, 1^q)$ family: $q = 0$ accidentally matches (recursion ill-defined), $q = 1$ fails (recursion valid, vanishing fails), and $q \geq 2$ matches structurally.

Mod-P verification at $P = 100\,003$, $Q = 11$ confirms all dim values listed.

1 Setup

We retain the notation of the May-13 multi-corner dimension paper. $H_q(S_n)$ is the Iwahori–Hecke algebra over $\mathbb{Q}(q)$ with generators $T_1, \dots, T_{n-1}$ satisfying $(T_i - q)(T_i + 1) = 0$ and braid. V_λ is the Specht module in the Hoefsmit seminormal basis $\{v_T\}_{T \in \mathrm{SYT}(\lambda)}$. Eigenspaces of T_j in V_λ: E^+_j, E^-_j.

For $1 \leq k \leq n$, set

$$R'_k \;:=\; (T_{k-1}{+}1)(T_{k-2}{+}1) \cdots (T_1{+}1) \;\in\; H_q(S_n), \qquad R'_1 := 1.$$

For $1 \leq a \leq b \leq n$, $P_{a,b} := R'_a R'_{a+1} \cdots R'_b$ (empty product is 1). The j-image at sharp index is

$$B^{(\lambda)}_{\ell+1} V_\lambda \;:=\; P_{\ell+1,n}\, V_\lambda \;=\; R'_{\ell+1} R'_{\ell+2} \cdots R'_n\, V_\lambda,$$

where $\ell = \ell(\lambda)$ and $j_0(\lambda) := \ell + 1$.

For a partition λ with r length-preserving corners $c_1, \dots, c_r$ and column-1 bottom $c_* = c_*(\lambda) = (\ell, 1)$ (when $\lambda_\ell = 1$), the corner-pair partitions are

$$\mu_X \;:=\; \lambda \setminus X, \qquad X \in \{\{c_i, c_j\} : i < j\} \cup \{\{c_i, c_*\}\}.$$

Throughout, $\widehat{\lambda}$ denotes the "head" of λ obtained by deleting trailing 1-rows.

1.1 Existing ingredients

We cite the toolkit verbatim:

Theorem 1 (Phase A endpoint, May-19)**.** *For any $H_q(S_n)$-module V, $R'_n V = E^+_{n-1}(V)$.*

Theorem 2 (Corner-pair embedding Φ_X, May-20 + SRD-421)**.** *For each corner pair $X = \{c, c'\}$ of λ with axial distance $|d_X| \geq 2$, there is a $\mathbb{Q}(q)$-linear injection $\Phi_X : V_{\mu_X} \hookrightarrow V_\lambda$ such that:*

1. *$\Phi_X(v_{T'})$ is the T_{n-1}-$(+q)$-eigenvector of the 2-block at the corner pair X where T' fills the rest.*
2. *$T_i\,\Phi_X(v) = \Phi_X(T_i v)$ for $1 \leq i \leq n-3$.*
3. *$\Phi_X(V_{\mu_X})$ is the span of T_{n-1}-$(+q)$-eigenvectors in the X 2-blocks.*
4. *For $n \geq 4$, Φ_X takes $E^-_1|_{V_{\mu_X}}$ injectively into $E^-_1|_{V_\lambda}$.*

Theorem 3 (Hook j-formula, Shift Lemma, May-12)**.** *For a hook $\mu = (k, 1^q)$ with $q \geq k$:*

$$B^{(\mu)}_{j_0(\mu)} V_\mu \subseteq E^-_1|_{V_\mu}, \qquad \dim B^{(\mu)}_{j_0(\mu)} V_\mu = 1.$$

Equivalently, with $\widehat{\mu} = (k)$, threshold $q_0(\widehat{\mu}) = k$ $(= |\widehat{\mu}| - 2\ell(\widehat{\mu}) + 2)$.

Theorem 4 (Two-column j-formula, May-19)**.** *For a two-column shape $\mu = (2^a, 1^q)$ with $q \geq a$:*

$$B^{(\mu)}_{j_0(\mu)} V_\mu \subseteq E^-_1|_{V_\mu}, \qquad \dim B^{(\mu)}_{j_0(\mu)} V_\mu = 1.$$

Lemma 5 (Two-column dim formula at $a = 2$, all $q \geq 0$)**.** *For $\mu = (2, 2, 1^q)$ with $q \geq 0$, $\dim B^{(\mu)}_{j_0(\mu)} V_\mu = 1$.*

Proof. Case $q \geq 1$ (recursive). μ has $r = 1$ length-preserving corner $c_1 = (2, 2)$ and column-1 bottom $c_* = (q+2, 1)$. The single corner-pair partition is $\mu_1 := \mu \setminus \{c_1, c_*\} = (2, 1^q)$, a hook with $\dim B^{(\mu_1)}_{j_0(\mu_1)} V_{\mu_1} = 1$.

No same-row pairs. At row 1: $\mu_1 = 2 \geq 2$; placing $\{n-1, n\}$ at $(1,1), (1,2)$ forces $(2,1) > n-1$, so $(2,1) = n$, but n already at $(1,2)$: contradiction. At row 2: similar contradiction with column-1 tail. Rows ≥ 3 have $\mu_i = 1$. Hence Phase A decomposition has only the one corner-pair piece: $E^+_{n-1}|_{V_\mu} = \Phi_{c_1, c_*}(V_{\mu_1})$.

Bad-pair vanishing not needed. With $r = 1$, there are no bad-pair contributions.

Axial distance. $|d_{c_1,c_*}| = |((q+2) - 1) - (2 - 2)| = q + 1 \geq 2$, so Φ_{c_1,c_*} is non-degenerate (Theorem 2).

Recursion. By the standard commutation lemma,

$$B^{(\mu)}_{j_0(\mu)} V_\mu \;=\; \mathcal{O}_\mu\, \Phi_{c_1,c_*}\big(B^{(\mu_1)}_{j_0(\mu_1)} V_{\mu_1}\big),$$

where $\mathcal{O}_\mu = (T_{j_0(\mu)-1}{+}1) \cdots (T_{n-2}{+}1)$. Φ-injectivity (Theorem 2(3)) and outer-chain adjacent-injectivity (May-13 dim2-331-structural) preserve dim 1.

Case $q = 0$ ($\mu = (2,2)$, $n = 4$). Sharp index $j_0 = 3$. $V_{(2,2)}$ has dim 2 with seminormal basis $\{v_{T_1}, v_{T_2}\}$ where T_1, T_2 swap the entry pattern at the $(1,2)$-vs-$(2,1)$ positions of $\{2,3\}$. The structural skeleton (May-13 evening) gives $B_3^{(2,2)} V_{(2,2)} \subseteq E_2^+|_{V_{(2,2)}}$. The axial distance for T_2 at $\{(1,2),(2,1)\}$ is $|d_2| = 2$, so the T_2-2-block on $V_{(2,2)}$ is non-degenerate; its $+q$-eigenspace $E_2^+|_{V_{(2,2)}}$ has dim 1 (half of the dim-2 block). Hence $\dim B \le 1$. The lower bound $\dim B \ge 1$ follows from $\Omega = R_3' R_4' \neq 0$ on $V_{(2,2)}$ (e.g. on the SYT vector not in E_1^-, which has dim ≥ 1 since $\dim V_{(2,2)} = 2 = \dim E_1^+ + \dim E_1^-$ with both summands 1-dim). Hence $\dim B = 1$. $\square$

Theorem 6 (General same-row-dies framework, SRD-421 Theorem 14)**.** *Let $\widetilde{\lambda} \vdash N$ admit a same-row pair at row i ($\widetilde{\lambda}_i \ge 2$), with sub-shape $\widetilde{\nu}_i := \widetilde{\lambda} \setminus \{(i, \widetilde{\lambda}_i - 1), (i, \widetilde{\lambda}_i)\}$, set $\widetilde{\ell} := \ell(\widetilde{\lambda})$, $\widetilde{j}_0 := \widetilde{\ell} + 1$, and let $\widetilde{S}_i$ be the corresponding same-row span. Assume $\widetilde{\lambda}_i \ge 3$ (so $\ell(\widetilde{\nu}_i) = \widetilde{\ell}$) and $B_{j_0(\widetilde{\nu}_i)}^{(\widetilde{\nu}_i)} V_{\widetilde{\nu}_i} \subseteq E_1^-|_{V_{\widetilde{\nu}_i}}$. Then*

$$R'_{\widetilde{j}_0}\, R'_{\widetilde{j}_0+1} \cdots R'_{N-1}\, \widetilde{S}_i \;=\; 0.$$

2 Theorem A: $(3, 2, 1^q)$ at $q \ge 1$

2.1 Reduction setup

Let $\lambda := (3, 2, 1^q)$ with $q \ge 1$, $n := q + 5$, $\ell := q + 2$, $j_0 := \ell + 1 = q + 3$.

Length-preserving corners: $c_1 = (1, 3)$, $c_2 = (2, 2)$. Column-1 bottom: $c_* = (q + 2, 1)$ (well-defined since $\lambda_\ell = 1$ at $q \ge 1$).

Corner-pair partitions:

$$\begin{aligned}
\mu_A &:= \lambda \setminus \{c_1, c_*\} = (2, 2, 1^{q-1}), & |\mu_A| &= q + 3, \quad \ell(\mu_A) = q + 1,\\
\mu_B &:= \lambda \setminus \{c_2, c_*\} = (3, 1^q), & |\mu_B| &= q + 3, \quad \ell(\mu_B) = q + 1,\\
\mu_C &:= \lambda \setminus \{c_1, c_2\} = (2, 1^{q+1}), & |\mu_C| &= q + 3, \quad \ell(\mu_C) = q + 2.
\end{aligned}$$

Same-row pair at row 1 is empty. Removing cells $(1,2), (1,3)$ from λ leaves $\nu_1 = (1, 2, 1^q)$, which is *not* a partition (row 1 has 1 cell, row 2 has 2 cells, violating the partition property). Equivalently: any SYT of λ with $\{n-1, n\}$ at $(1,2), (1,3)$ would force the cell $(2,2)$ to hold a letter $\le n-2$ greater than the letter at $(2,1)$ but also less than $n-1$, simultaneously the only letter remaining at $(1,1)$ is 1; checking column 1 monotonicity from $(1,1)$ down forces the column-1 chain to be valid only if the two cells at row 2 have letters in $\{2, \ldots, n-2\}$, which is fine—but the SYT structure with $(1,1) = 1$, $(2,2) =$ small, $(2,1) =$ even smaller, fails because $(2,2) > (1,2) = n-1$ is impossible. We conclude no SYT realises a same-row pair at row 1.

Rows ≥ 2 have $\lambda_i \le 2$ with column-1 tail, so no same-row pair (the argument is the same as in (4,3,1)-UB and the (5,2,1) paper).

Hence the Phase A decomposition has *three* corner-pair pieces and *no* same-row part:

$$E_{n-1}^+|_{V_\lambda} \;=\; \Phi_A(V_{\mu_A}) \,\oplus\, \Phi_B(V_{\mu_B}) \,\oplus\, \Phi_C(V_{\mu_C}). \tag{1}$$

Axial distances. $d_A = |c_1 - c_*|_{\text{axial}} = q + 5 \ge 6$ ✓; $d_B = q + 1 \ge 2$ ✓; $d_C = 2$ ✓. All three pairs are non-degenerate. The decomposition (1) is direct (disjoint seminormal supports).

2.2 Commutation lemma

Lemma 7 (Commutation, May-12 / May-20)**.** *For each $X \in \{A, B, C\}$ and every $j_0 \le k \le n-1$, $R'_k\, \Phi_X(v) = (T_{k-1}{+}1)\, \Phi_X(R'^{(\mu_X)}_{k-1} v)$ for $v \in V_{\mu_X}$. Iterating,*

$$P_{j_0,n-1}\, \Phi_X(V_{\mu_X}) \;=\; \mathcal{O}_\lambda\, \Phi_X\big(P^{(\mu_X)}_{j_0-1,n-2}\, V_{\mu_X}\big), \tag{2}$$

where $\mathcal{O}_\lambda := (T_{j_0-1}{+}1)(T_{j_0}{+}1)\cdots(T_{n-2}{+}1)$.

2.3 Each piece evaluated

$\mu_A = (2, 2, 1^{q-1})$ **piece.** $j_0(\mu_A) = \ell(\mu_A) + 1 = q + 2 = j_0 - 1$. So $P^{(\mu_A)}_{j_0-1,n-2} V_{\mu_A} = B^{(\mu_A)}_{j_0(\mu_A)} V_{\mu_A}$, at the sharp index for μ_A. By Lemma 5 (two-column $a = 2$, all $q' \ge 0$), applied at $q' = q - 1 \ge 0$: $\dim B^{(\mu_A)}_{j_0(\mu_A)} V_{\mu_A} = 1$.

By Theorem 2(3), $\Phi_A(B^{(\mu_A)} V_{\mu_A})$ is a 1-dim subspace of $E^+_{n-1}|_{V_\lambda}$.

$\mu_B = (3, 1^q)$ **piece.** Hook with $j_0(\mu_B) = q + 2 = j_0 - 1$. Hook dim is 1: $\dim B^{(\mu_B)}_{j_0(\mu_B)} V_{\mu_B} = 1$.

By Theorem 2(3), $\Phi_B(B^{(\mu_B)} V_{\mu_B})$ is a 1-dim subspace of $E^+_{n-1}|_{V_\lambda}$.

$\mu_C = (2, 1^{q+1})$ **piece: leakage to zero.** Hook with $j_0(\mu_C) = q + 3 = j_0$, one step *above* $j_0 - 1$. So

$$P^{(\mu_C)}_{j_0-1,n-2} V_{\mu_C} \;=\; R'^{(\mu_C)}_{j_0-1} \cdot B^{(\mu_C)}_{j_0(\mu_C)} V_{\mu_C}.$$

By Theorem 3 (hook j-formula on $(2, 1^{q+1})$) at $q + 1 \ge 2$, i.e. $q \ge 1$: $B^{(\mu_C)}_{j_0(\mu_C)} V_{\mu_C} \subseteq E^-_1|_{V_{\mu_C}}$. The operator $R'^{(\mu_C)}_{j_0-1}$ has trailing factor $(T_1{+}1)$, which kills $E^-_1|_{V_{\mu_C}}$. Hence $P^{(\mu_C)}_{j_0-1,n-2} V_{\mu_C} = 0$, and the C-piece vanishes.

2.4 The upper bound

Lemma 8 (Upper bound for $(3, 2, 1^q)$)**.** *For $\lambda = (3, 2, 1^q)$ at every $q \ge 1$, $\dim B^{(\lambda)}_{j_0} V_\lambda \le 2$.*

Proof. By (1) and Theorem 1, $B^{(\lambda)}_{j_0} V_\lambda = P_{j_0,n-1} R'_n V_\lambda = P_{j_0,n-1} E^+_{n-1}|_{V_\lambda}$. Apply $P_{j_0,n-1}$ to each summand of (1) and use Lemma 7:

$$B^{(\lambda)}_{j_0} V_\lambda \;=\; \mathcal{O}_\lambda \Big[\sum_{X \in \{A,B,C\}} \Phi_X\big(P^{(\mu_X)}_{j_0-1,n-2}\, V_{\mu_X}\big) \Big].$$

The C-piece vanishes outright (above), so

$$B^{(\lambda)}_{j_0} V_\lambda \;=\; \mathcal{O}_\lambda \Big[\Phi_A\big(B^{(\mu_A)} V_{\mu_A}\big) \;+\; \Phi_B\big(B^{(\mu_B)} V_{\mu_B}\big) \Big].$$

By the dimensions above, each summand inside is $\le$ 1-dim. Hence $\dim B^{(\lambda)}_{j_0} V_\lambda \le 1 + 1 = 2$. $\square$

2.5 Lower bound

Lemma 9 (Disjoint Φ_A, Φ_B supports)**.** *$\Phi_A(V_{\mu_A}) \cap \Phi_B(V_{\mu_B}) = 0$ in V_λ.*

Proof. $\Phi_X(v_{T'})$ is supported on SYTs of λ placing $\{n-1, n\}$ at the cell pair X. The unordered cell pairs are $A = \{(1,3),(q+2,1)\}$ and $B = \{(2,2),(q+2,1)\}$, which share c_* but differ in the other element ($c_1 \neq c_2$). Hence the seminormal supports of $\Phi_A(V_{\mu_A})$ and $\Phi_B(V_{\mu_B})$ are disjoint, and the intersection is 0. □

Theorem 10 (Adjacent-injectivity of $\mathcal{O}_\lambda$, May-13 dim2-331-structural)**.** *The outer chain $\mathcal{O}_\lambda = (T_{j_0-1}+1)\cdots(T_{n-2}+1)$ is injective on $E^+_{n-1}|_{V_\lambda}$.*

Lemma 11 (Lower bound)**.** *For $\lambda = (3,2,1^q)$ at every $q \geq 1$, $\dim B^{(\lambda)}_{j_0} V_\lambda \geq 2$.*

Proof. $\Phi_A(B^{(\mu_A)}) \oplus \Phi_B(B^{(\mu_B)})$ has dimension $1+1=2$ by Lemma 9. Both summands lie in $E^+_{n-1}|_{V_\lambda}$ (Theorem 2(3)). The outer chain $\mathcal{O}_\lambda$ is injective on $E^+_{n-1}|_{V_\lambda}$ by Theorem 10. Hence $\dim B^{(\lambda)}_{j_0} V_\lambda \geq 2$. □

Theorem 12 (Theorem A)**.** *For $\lambda = (3,2,1^q)$ at every $q \geq 1$,*

$$\dim B^{(\lambda)}_{j_0} V_\lambda \;=\; 2 \;=\; f^{(2,1)} \;=\; f^{\lambda(\geq 2)}.$$

Proof. Combine Lemmas 8 and 11. □

3 Theorem B: $(4,2,1^q)$ at $q \geq 2$

3.1 Reduction setup

Let $\lambda := (4,2,1^q)$ with $q \geq 2$, $n := q+6$, $\ell := q+2$, $j_0 := q+3$.

Length-preserving corners: $c_1 = (1,4)$, $c_2 = (2,2)$; $c_* = (q+2,1)$.

Corner-pair partitions:

$$\begin{aligned}
\mu_A &:= \lambda \setminus \{c_1, c_*\} = (3,2,1^{q-1}), && |\mu_A| = q+4, \quad \ell(\mu_A) = q+1,\\
\mu_B &:= \lambda \setminus \{c_2, c_*\} = (4,1^q), && |\mu_B| = q+4, \quad \ell(\mu_B) = q+1,\\
\mu_C &:= \lambda \setminus \{c_1, c_2\} = (3,1^{q+1}), && |\mu_C| = q+4, \quad \ell(\mu_C) = q+2.
\end{aligned}$$

Same-row pair at row 1 realised. $\lambda_1 = 4 \geq 2$, cells $(1,3),(1,4)$. Sub-shape $\nu_1 := \lambda \setminus \{(1,3),(1,4)\} = (2,2,1^q)$, a valid partition with $\ell(\nu_1) = q+2 = \ell(\lambda)$ and $j_0(\nu_1) = q+3 = j_0(\lambda)$. The same-row pair at row 1 is realised.

Row 2: $\lambda_2 = 2$, same argument as in (5,2,1) paper shows no same-row pair.

Phase A decomposition (four pieces):

$$E^+_{n-1}|_{V_\lambda} \;=\; \Phi_A(V_{\mu_A}) \,\oplus\, \Phi_B(V_{\mu_B}) \,\oplus\, \Phi_C(V_{\mu_C}) \,\oplus\, S_1, \tag{3}$$

where $S_1 = \operatorname{span}\{v_T : T(1,3) = n-1, T(1,4) = n\}$ is the same-row part at row 1.

Axial distances. $d_A = q+5 \geq 7$; $d_B = q+1 \geq 3$; $d_C = 3$. All non-degenerate.

3.2 Each piece evaluated at $q \geq 2$

$\mu_A = (3, 2, 1^{q-1})$ piece. $j_0(\mu_A) = q + 2 = j_0 - 1$. By Theorem 12 (this paper!) at $q - 1 \geq 1$, i.e. $q \geq 2$: $\dim B^{(\mu_A)}_{j_0(\mu_A)} V_{\mu_A} = 2$.

Hence $\Phi_A(B^{(\mu_A)} V_{\mu_A})$ is 2-dim in $E^+_{n-1}|_{V_\lambda}$.

$\mu_B = (4, 1^q)$ piece. Hook with $j_0(\mu_B) = q + 2 = j_0 - 1$. $\dim B^{(\mu_B)}_{j_0(\mu_B)} V_{\mu_B} = 1$.

$\mu_C = (3, 1^{q+1})$ piece: leakage to zero. Hook with $j_0(\mu_C) = q + 3 = j_0$. So $P^{(\mu_C)}_{j_0-1,n-2} V_{\mu_C} = R'^{(\mu_C)}_{j_0-1} \cdot B^{(\mu_C)}_{j_0(\mu_C)} V_{\mu_C}$. By Theorem 3 (hook on $(3, 1^{q+1})$) at $q+1 \geq 3$, i.e. $q \geq 2$: $B^{(\mu_C)}_{j_0(\mu_C)} V_{\mu_C} \subseteq E^-_1|_{V_{\mu_C}}$. $R'^{(\mu_C)}_{j_0-1}$ ends with (T_1+1), killing E^-_1. Hence $P^{(\mu_C)}_{j_0-1,n-2} V_{\mu_C} = 0$ and the C-piece vanishes.

Same-row S_1 piece: dies via SRD framework. Sub-shape $\nu_1 = (2, 2, 1^q)$, two-column with $a = 2$ and q trailing 1-rows. By Theorem 4 at $q \geq 2$: $B^{(\nu_1)}_{j_0(\nu_1)} V_{\nu_1} \subseteq E^-_1|_{V_{\nu_1}}$. Apply Theorem 6 with $\widetilde{\lambda} = \lambda$, $i = 1$, $\widetilde{\lambda}_i = 4 \geq 3$, $\ell(\nu_1) = \ell(\lambda)$:

$$R'_{j_0}\, R'_{j_0+1} \,\cdots\, R'_{n-1}\, S_1 \;=\; 0.$$

Hence the S_1-piece of (3) vanishes after $P_{j_0,n-1}$.

3.3 Upper and lower bounds

Lemma 13 (Upper bound for $(4, 2, 1^q)$)**.** *For $\lambda = (4, 2, 1^q)$ at every $q \geq 2$, $\dim B^{(\lambda)}_{j_0} V_\lambda \leq 3$.*

Proof. By (3), Theorem 1, and the commutation lemma,

$$B^{(\lambda)}_{j_0} V_\lambda \;=\; \mathcal{O}_\lambda \Big[\Phi_A\big(B^{(\mu_A)}\big) + \Phi_B\big(B^{(\mu_B)}\big) + \Phi_C\big(P^{(\mu_C)}_{j_0-1,n-2} V_{\mu_C}\big) \Big] + P_{j_0,n-1}\, S_1.$$

The C-piece and S_1-piece vanish at $q \geq 2$ by Sections 3.2 (above). Hence

$$\dim B^{(\lambda)}_{j_0} V_\lambda \;\leq\; \dim \Phi_A(B^{(\mu_A)}) + \dim \Phi_B(B^{(\mu_B)}) \;=\; 2 + 1 \;=\; 3.$$

□

Lemma 14 (Disjoint Φ_A, Φ_B supports for $(4, 2, 1^q)$)**.** $\Phi_A(V_{\mu_A}) \cap \Phi_B(V_{\mu_B}) = 0$ *in* V_λ.

Proof. A-pair $\{c_1, c_*\} = \{(1, 4), (q + 2, 1)\}$, B-pair $\{c_2, c_*\} = \{(2, 2), (q + 2, 1)\}$. They share c_* but differ in the other element. Same argument as Lemma 9. □

Lemma 15 (Lower bound)**.** *For $\lambda = (4, 2, 1^q)$ at every $q \geq 2$, $\dim B^{(\lambda)}_{j_0} V_\lambda \geq 3$.*

Proof. $\Phi_A(B^{(\mu_A)}) \oplus \Phi_B(B^{(\mu_B)})$ has dimension $2 + 1 = 3$ by Lemma 14. Both lie in $E^+_{n-1}|_{V_\lambda}$. Outer chain $\mathcal{O}_\lambda$ is injective on $E^+_{n-1}|_{V_\lambda}$ (Theorem 10). Hence $\dim \geq 3$. □

Theorem 16 (Theorem B)**.** *For $\lambda = (4, 2, 1^q)$ at every $q \geq 2$,*

$$\dim B^{(\lambda)}_{j_0} V_\lambda \;=\; 3 \;=\; f^{(3,1)} \;=\; f^{\lambda(\geq 2)}.$$

4 The non-stable failures: $q = 0, 1$

4.1 $\lambda = (4, 2)$, $q = 0$: recursion ill-defined

For $\lambda = (4, 2)$, $\ell = 2$ but $\lambda_\ell = 2 \neq 1$. The candidate column-1 bottom $c_* = (2, 1)$ is *not* removable (removing it leaves the cell $(2, 2)$ alone at row 2, which is not a valid partition). Hence the four-branch recursion does not apply at $\lambda = (4, 2)$.

Direct computation gives $\dim B_3^{(4,2)} V_{(4,2)} = 3 = f^{(3,1)}$. The numerical match is an *accident*; no structural argument here derives it. The closed-form formula is correct at $q = 0$ as a combinatorial coincidence.

4.2 $\lambda = (4, 2, 1)$, $q = 1$: vanishing arguments fail

At $q = 1$, $\lambda = (4, 2, 1)$, $n = 7$, $\ell = 3$, $j_0 = 4$. Direct computation: $\dim B_4^{(\lambda)} V_\lambda = 6 \neq 3$.

The four-branch decomposition (3) applies, but the vanishing arguments *fail*:

$\mu_C = (3, 1^2)$ **leakage fails.** μ_C has $q(\mu_C) = 2$, hook threshold for $\widehat{\mu_C} = (3)$ is $q + 1 \geq 3$, i.e. $q(\mu_C) \geq 3$. At $q(\mu_C) = 2$, we have $B_{j_0(\mu_C)}^{(\mu_C)} V_{\mu_C} \not\subseteq E_1^- |_{V_{\mu_C}}$ (verified computationally). Hence $R'^{(\mu_C)}_{j_0 - 1} B_{j_0(\mu_C)}^{(\mu_C)} V_{\mu_C} \neq 0$. Direct computation: $\dim P_{j_0-1,n-2}^{(\mu_C)} V_{\mu_C} = 1$.

$\nu_1 = (2, 2, 1)$ **same-row leakage fails.** ν_1 has $\widehat{\nu_1} = (2, 2)$, two-column threshold for ν_1 is $q(\nu_1) \geq a = 2$. At $q(\nu_1) = 1$, we have $B_{j_0(\nu_1)}^{(\nu_1)} V_{\nu_1} \not\subseteq E_1^- |_{V_{\nu_1}}$. The SRD framework does not apply, and $P_{j_0,n-1} S_1 \neq 0$.

Total dimension at $q = 1$.

$$\dim B_{j_0}^{(\lambda)} V_\lambda \; = \; \underbrace{2+1}_{\text{good pairs}} \; + \; \underbrace{1}_{\text{bad pair } \mu_C} \; + \; \underbrace{2}_{\text{same-row } S_1} \; = \; 6.$$

The 2-dim same-row contribution is established by direct computation ($\dim P_{j_0,n-1} S_1 = 2$ at $q = 1$). The bad-pair contribution is 1 (above). The good-pair contributions remain $2 + 1 = 3$.

The non-monotonicity $\dim_{q=0} = 3$, $\dim_{q=1} = 6$, $\dim_{q=2} = 3$ is therefore fully explained:

- $q = 0$: recursion does not apply, but $\dim = 3$ matches by coincidence.
- $q = 1$: recursion applies, but bad-pair and same-row branches do not vanish, adding $1 + 2 = 3$ to the good-pair total of 3.
- $q \geq 2$: recursion applies, all vanishing arguments work, giving the closed form $\dim = 3$.

5 Computational verification

Mod $P = 100\,003$ at $Q = 11$:

$(3, 2, 1^q)$ **family.**

q	n	$\dim V$	$\dim B$
0	5	5	2
1	6	16	2
2	7	35	2
3	8	64	2
4	9	105	2

Matches Theorem 12 at $q \geq 1$ structurally. The $q = 0$ value dim $= 2$ is a numerical fact not derivable from the recursion (which doesn't apply).

$(4, 2, 1^q)$ **family.**

q	n	$\dim V$	$\dim B$
0	6	9	3 (accidental)
1	7	35	**6** (extra 3)
2	8	90	3
3	9	189	3
4	10	350	3

Matches Theorem 16 at $q \geq 2$ structurally. The $q = 0$ match is accidental; the $q = 1$ failure is dissected in Section 4.

Script. `~/projects/scratch/2026-05-13-prove-iso/threshold_scan.py`, `dissect_421_q1.py`.

6 Honest scope

What is proved structurally. Theorems 12, 16: the dim formula $\dim B^{(\lambda)}_{\ell+1} V_\lambda = f^{\lambda^{(\geq 2)}}$ holds for $\lambda \in \{(3, 2, 1^q)\}_{q\geq 1} \cup \{(4, 2, 1^q)\}_{q\geq 2}$.

What this extends. The existing structural-dim-formula club required $n \geq 8$ for $(3, 2, 1^q)$ (May-20) and $n \geq 10$ for $(4, 2, 1^q)$ (May-12 Eve3). This paper closes those down to $n \geq 6$ and $n \geq 8$ respectively, matching the computational table of the May-13 multi-corner dimension paper across $q \in \{1, 2\}$ structurally.

What is conditional. None of the structural inputs are open: Phase A endpoint (May-19), Φ_X embedding (May-20), hook j-formula (Shift Lemma), two-column j-formula (May-19), SRD framework (SRD-421), adjacent-injectivity (May-13 dim2-331-structural), and Theorem 12 (this paper, used in Theorem 16).

What is non-stable. At $q = 0$ for $(4, 2)$, the recursion does not apply; the dim equality holds by coincidence. At $q = 1$ for $(4, 2, 1)$, the recursion applies but vanishing fails; the dim is 6, not 3. Section 4 explains both phenomena structurally.

Connection to threshold conjecture. The data and the structural proofs suggest the *dim threshold* for the family $\lambda = (\widehat{\lambda}, 1^q)$ is

$$q_0^{\dim}(\widehat{\lambda}) \; = \; \begin{cases} 0, & |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) \le 1, \\ |\widehat{\lambda}| - 2\ell(\widehat{\lambda}), & |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) \ge 2. \end{cases}$$

This is one less than the night $T_1 = -1$ threshold $q_0(\widehat{\lambda}) = |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2$. The dim equality holds at $q \ge q_0^{\dim}$, even when $T_1 = -1$ fails on B. The verified data:

$\widehat{\lambda}$	$\lvert\widehat{\lambda}\rvert - 2\ell(\widehat{\lambda})$	$q_0^{\dim}$ predicted	dim matches at $q \ge$
$(3,2)$	1	0	0 (Theorem 12 at $q \ge 1$; $q = 0$ checked computationally)
$(3,2,2)$	1	0	0 (computational; structural proof open)
$(4,2)$	2	2	2 (Theorem 16); $q = 0$ accidental match
$(3,3)$	2	2	2 (computational; structural proof in May-13 SRD-331 + recursion)
$(4,3)$	3	3	3 (computational + May-13 afternoon structural at $q \ge 5$ via $T_1 = -1$)
$(5,2)$	3	3	3 (computational + May-13 late-night structural at $q \ge 5$ via $T_1 = -1$)

The structural proof of $q_0^{\dim} \le |\widehat{\lambda}| - 2\ell(\widehat{\lambda})$ (rather than the higher threshold $|\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2$ for $T_1 = -1$) is the new content of this paper, established for the two specific families.

What is next. A general structural proof of the dim threshold conjecture $q_0^{\dim}(\widehat{\lambda}) = \max(0, |\widehat{\lambda}| - 2\ell(\widehat{\lambda}))$ would require extending the four-branch reduction to arbitrary $\widehat{\lambda}$ at $q \ge q_0^{\dim}$, and recursively closing the sub-shape inputs (μ_A, ν_1, etc. at $q' \ge q_0^{\dim}(\widehat{\mu_A})$, etc.). The chained reduction works whenever each sub-shape's threshold condition descends along the recursion. Verifying this for all multi-corner shapes is the natural next target.